

PARTS LIST

Semiconductors:

- IC1 - PIC16F877A microcontroller
- IC2 - 7805, 5V regulator
- BR1, BR2 - 1A bridge rectifier module
- LCD1 - 16x2 LCD
- D1-D5 - 1N4007 rectifier diode
- T1-T5 - BC548 npn transistor
- ZD1, ZD2 - 5.1V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3, R5, R7-R9 - 10-kilo-ohm
- R2 - 100-ohm
- R4 - 330-ohm
- R6 - 47-kilo-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:

- C1, C7 - 0.1 μ F ceramic
- C2, C3 - 22pF ceramic
- C4, C5 - 10 μ F, 25V electrolytic
- C6 - 10 μ F, 35V electrolytic

Miscellaneous:

- CON1-CON4 - 2-pin terminal connector
- RL1-RL3 - 12V, 1 C/O relay
- S1 - Tactile switch
- X1 - 230V AC primary to 9V, 300mA secondary
- X2 - Current transformer, 50/1A ratio
- X_{TAL}1 - 4MHz crystal oscillator

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

ponents and double-check for any overlooked error. Use proper IC base for the microcontroller.

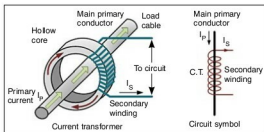


Fig. 5: Current transformer

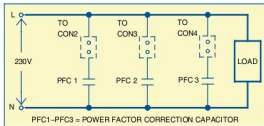


Fig. 6: Block diagram of PFC capacitor connections

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	12V
TP2	5V
TP3	Low (when S1 is pressed)
TP4	High pulse (voltage zero-crossing detector)
TP5	High pulse (current zero-crossing detector)

Connect transformer X1 to the same AC mains supply where the load (motor) is connected. Connect the secondary terminals of current transformer X2 to the circuit. Power supply cable of the load should pass through the current transformer as shown in Fig. 5.

When you power on the circuit, the LCD will show the original or existing power factor value. You can find out the value of PFC capacitor as indicated above. The PFC capacitors and the load are to be connected externally. The PFC capacitor is connected across the load through the relay contact as shown in Fig. 6. Extend two-wire cable from the dotted line, as shown in the block diagram, to the respective connector on the PCB.

To test the circuit for proper functioning, verify the correct 12V and 5V supplies in the circuit at TP1 and TP2 with respect to TP0. Refer test points Table II for checking at various points in the circuit. ●

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